

Underwater Acoustic Direction Finding

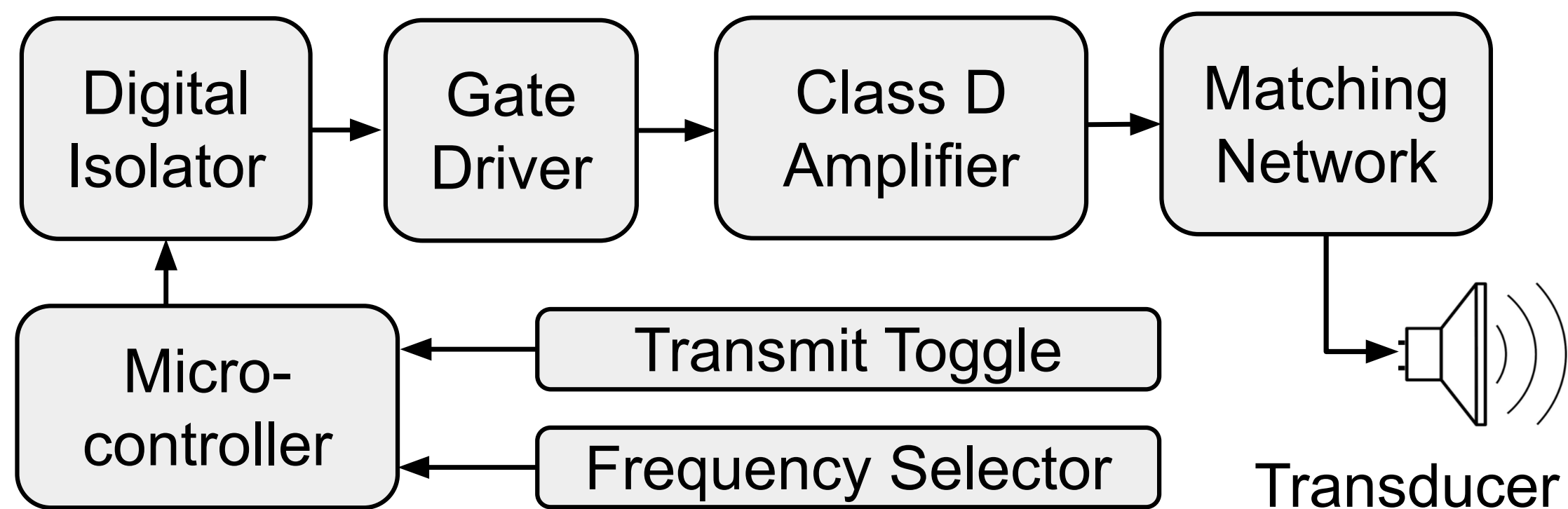
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Date of Completion : 4/10/2026

System Overview

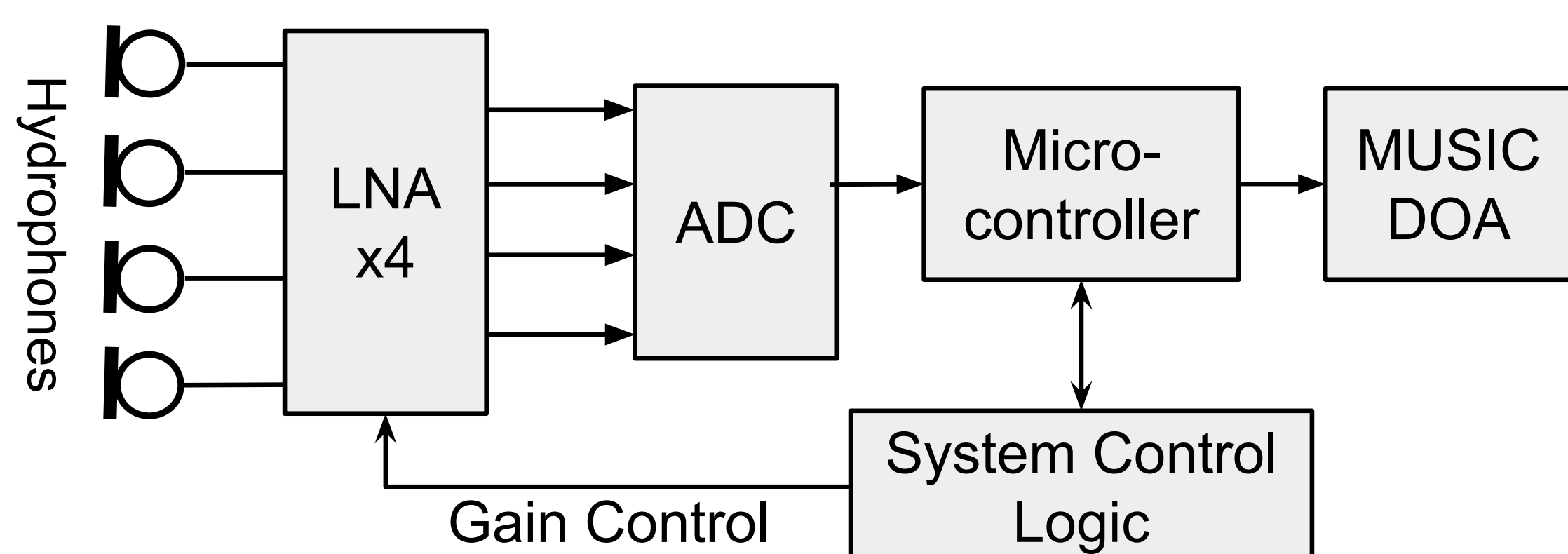
Transmitter

Transmit 25 - 40 kHz acoustic signals that are strong enough to travel through a pool

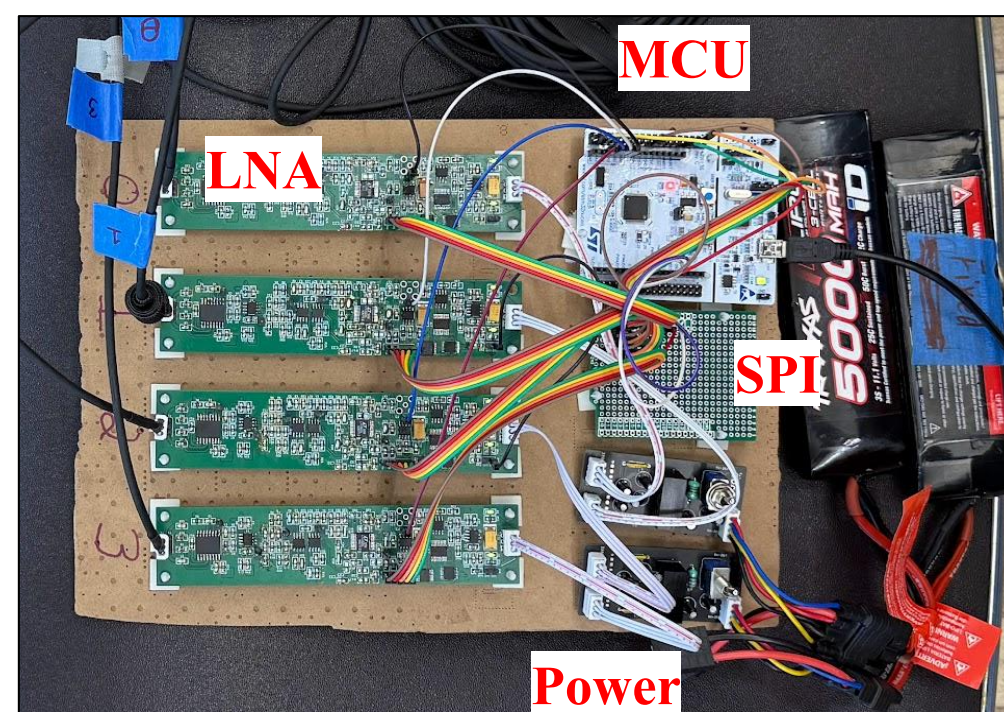
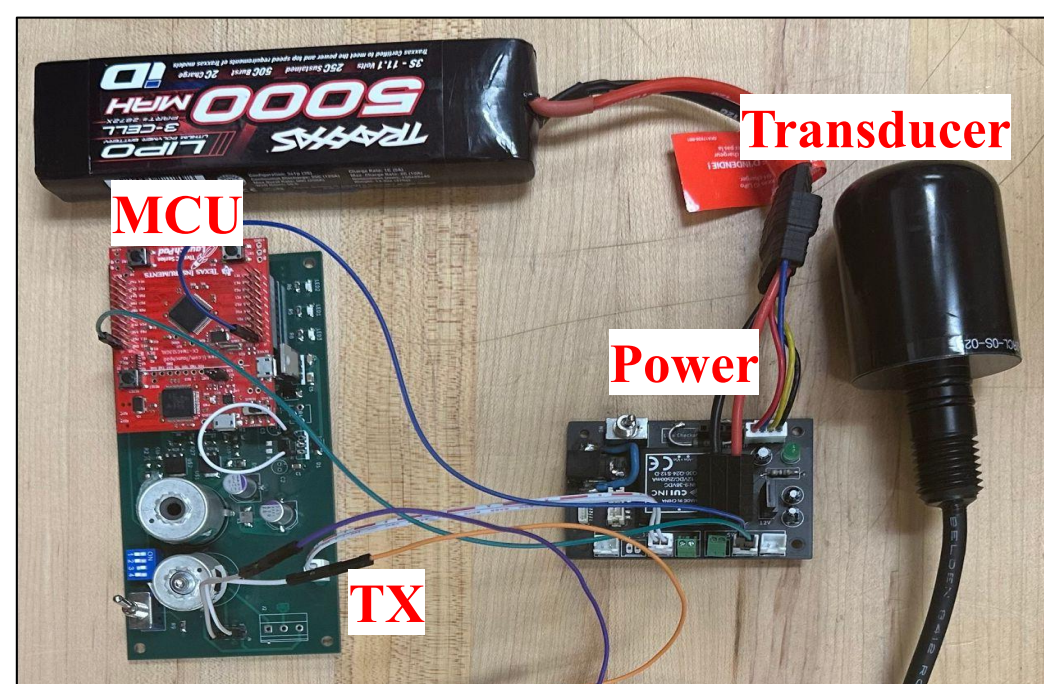


Receiver

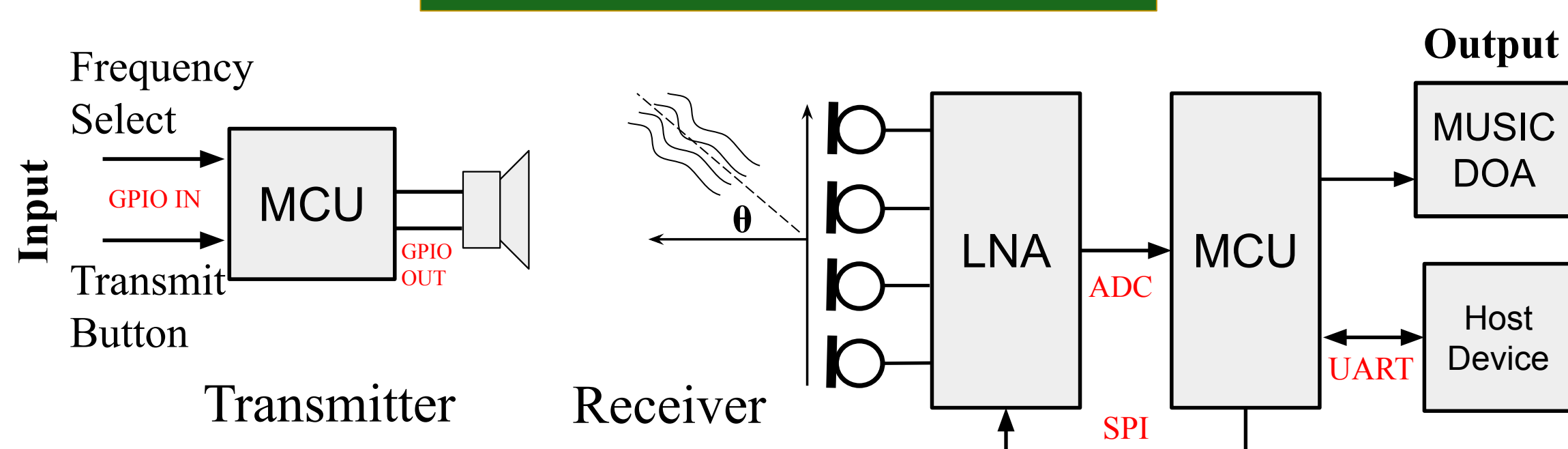
Perform DOA calculations on data received from transmitter as it arrives at a 4 element array



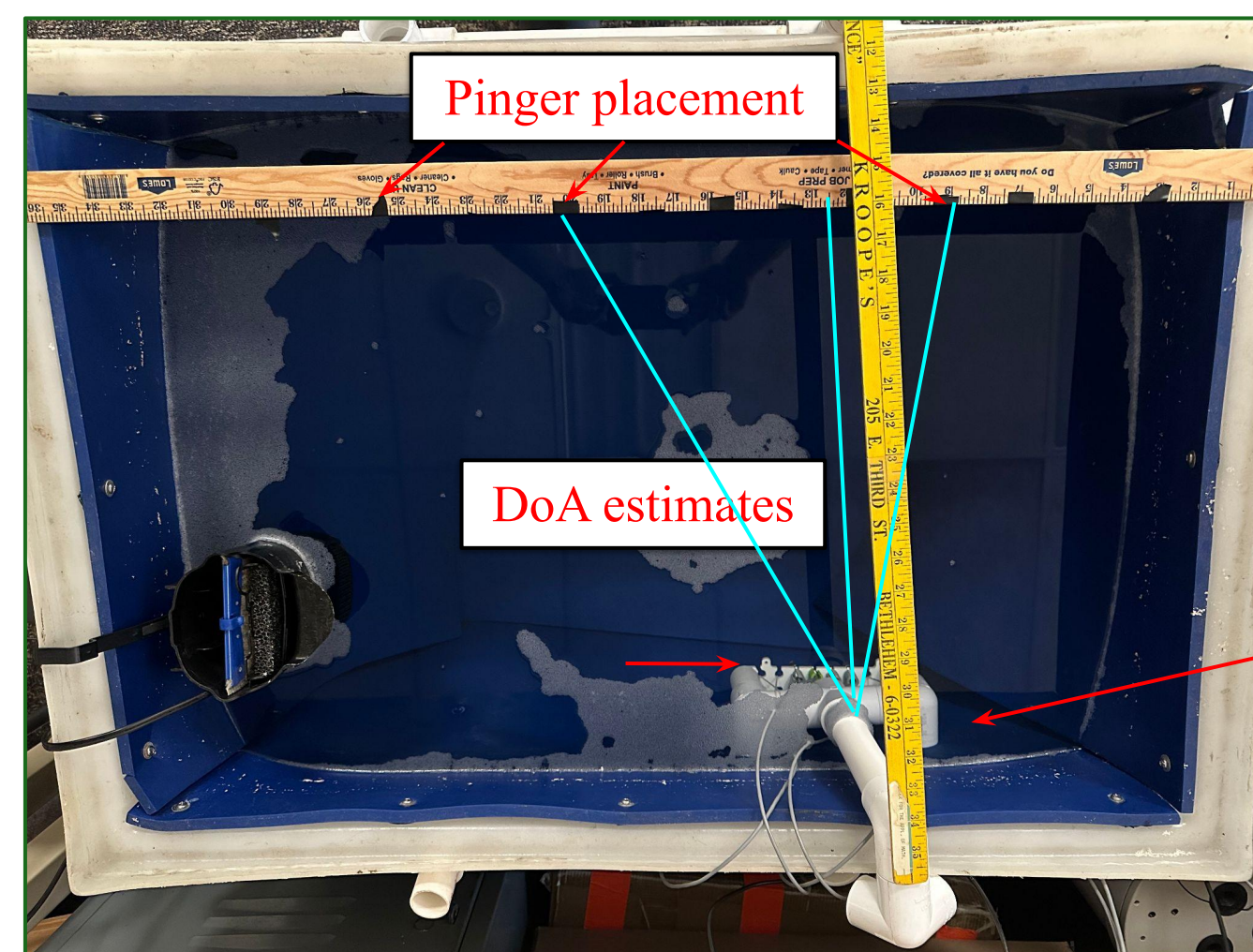
Hardware



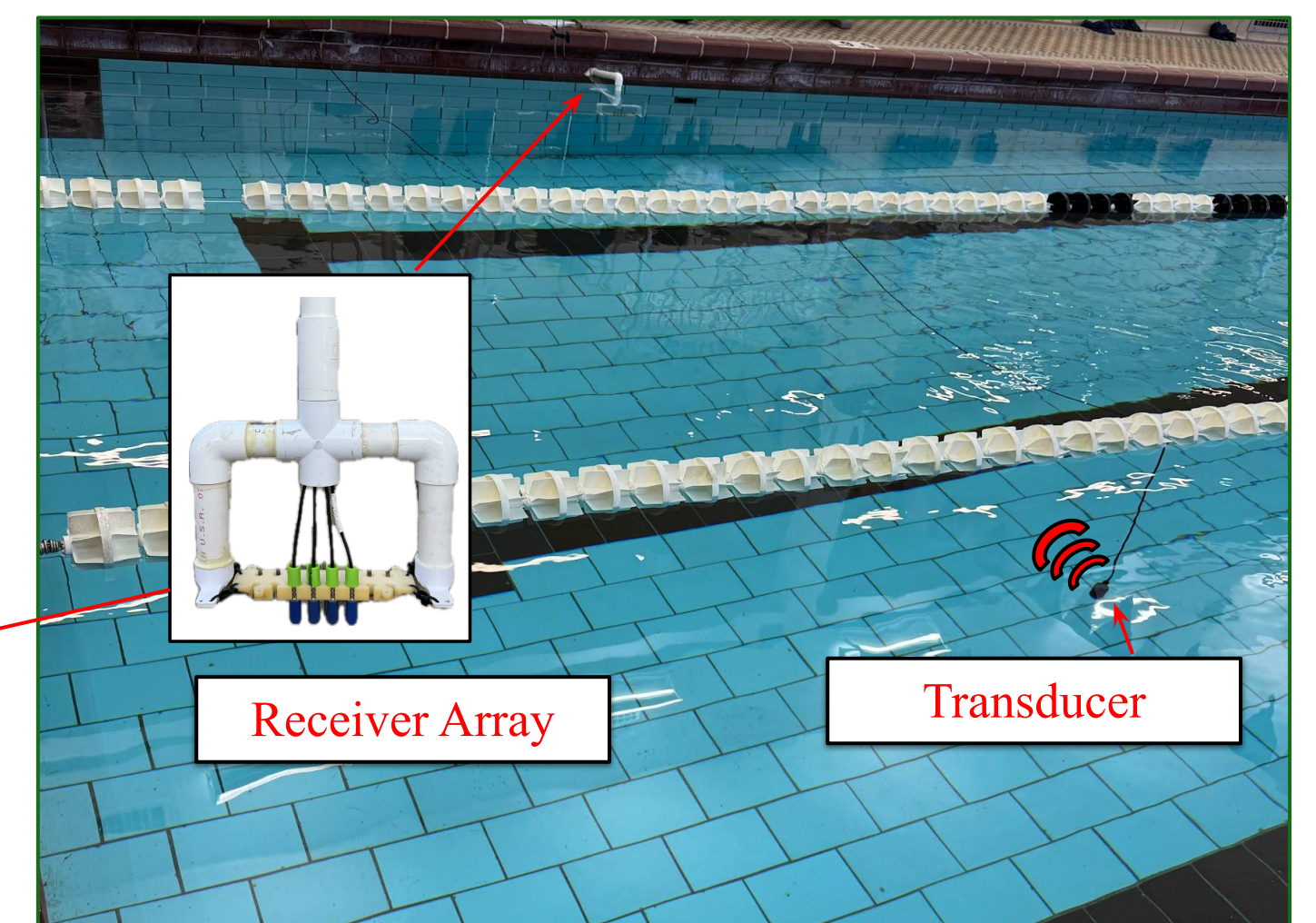
Integration



Experimental Setup

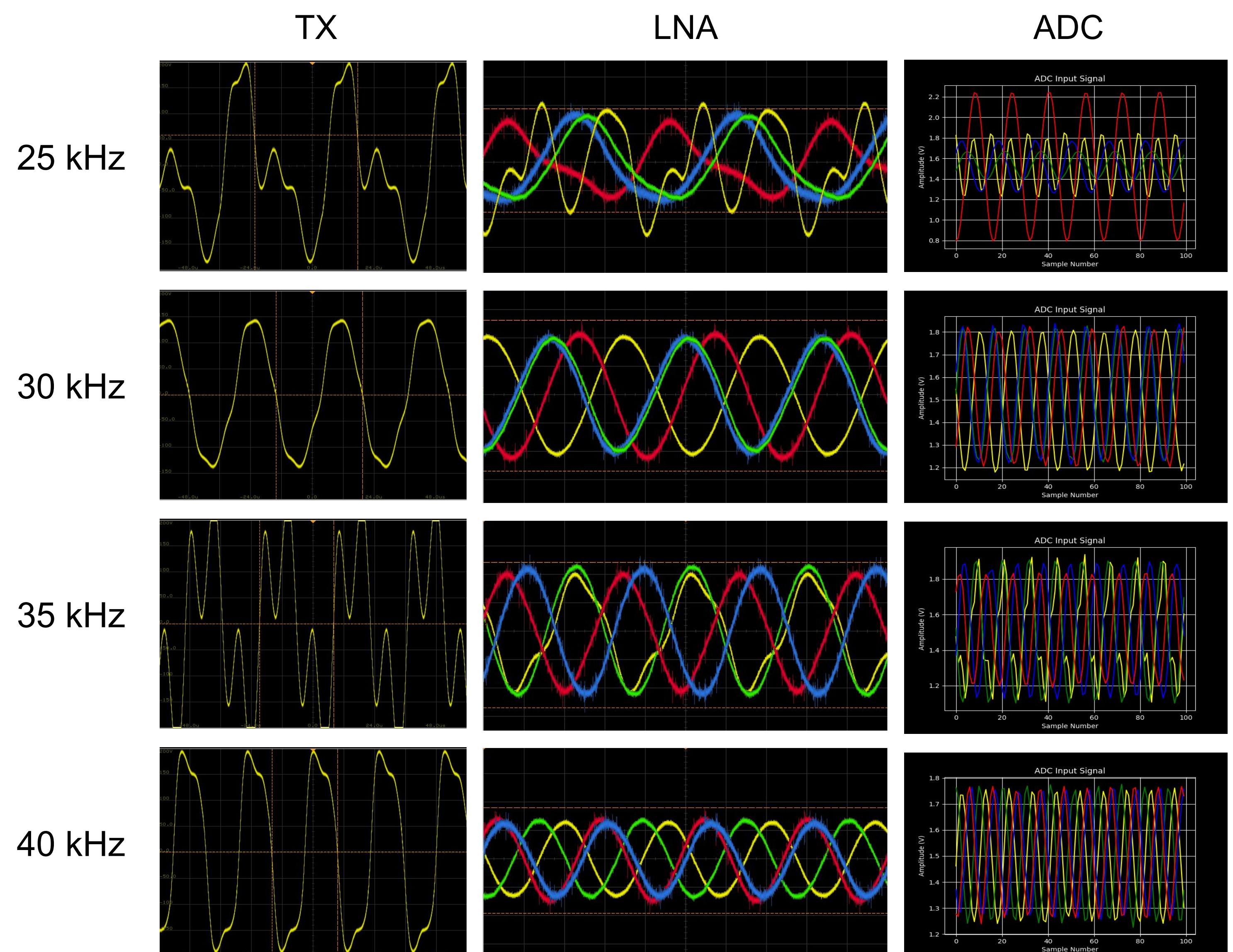


Anechoic Water Tank Test



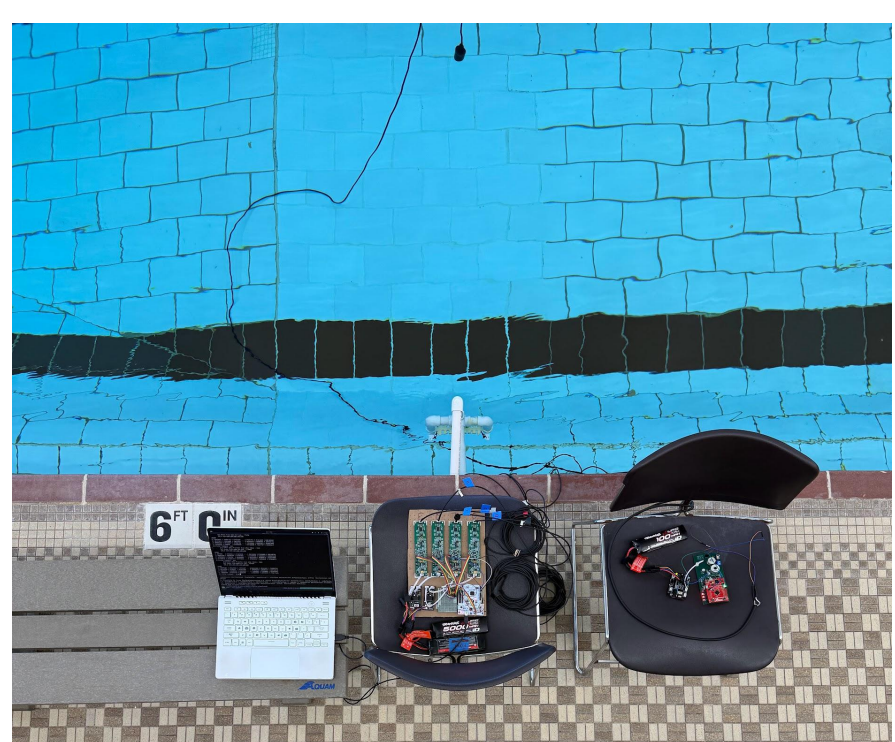
Swimming Pool Test

Results and Performance

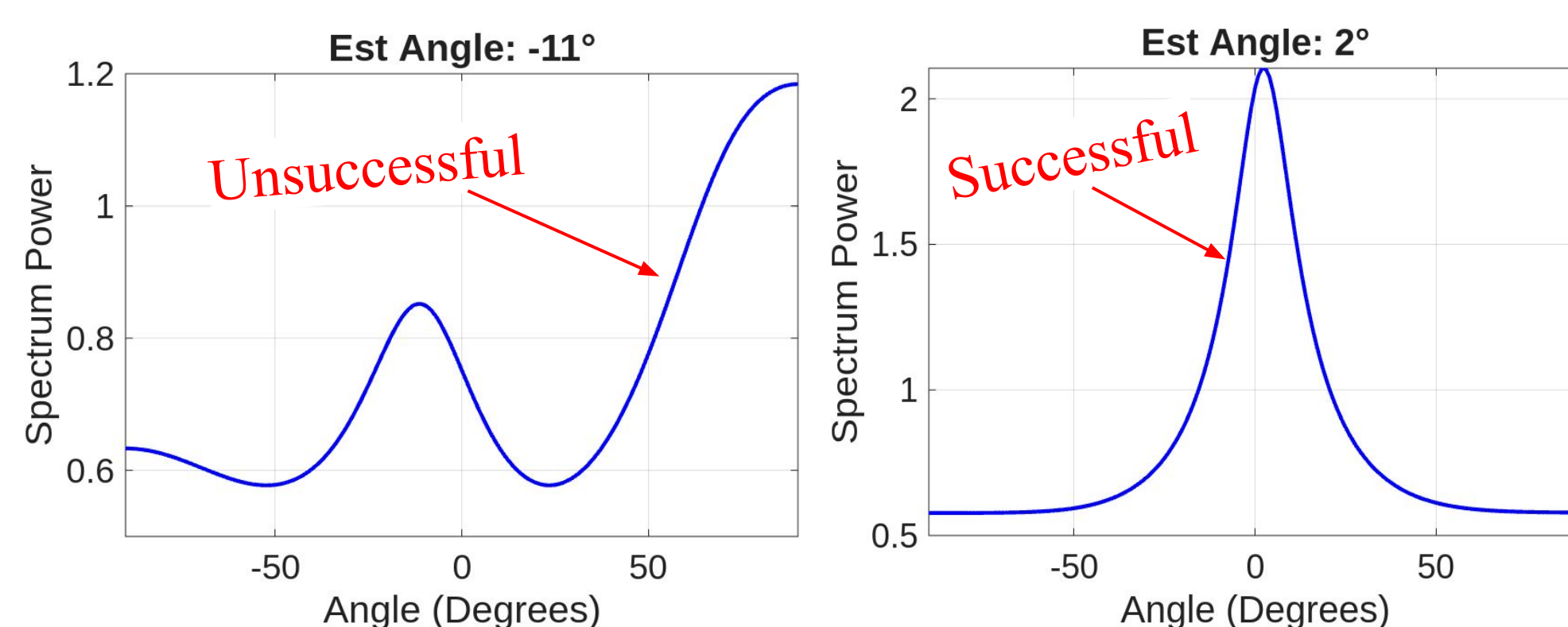


- ADC data is filtered and gain can be controlled using SPI to adjust signal amplitude
- Can reliably produce and capture clean signals onto an STM32 MCU
- Direction estimation on ADC data is inconsistent, but can produce results within +/- 10 degrees
- ~4.5 ms from initiating a DOA conversion and receiving the result on the host computer

System Demonstration



Pool Test Configuration



DOA estimation of sample

Limitations

- Echos in pool environment create correlated signals, degrading DOA estimation
- LNA board size is prohibitive to drone integration

Conclusion

- Can transmit 25-40kHz acoustic signals
- Receiver system can capture ADC data and perform DOA calculations using a command based communication protocol
- DOA calculations are accurate on simulated data, and have been shown to work on collected data